



9.3.2 The Power Method: Convergence ¶

9.3.2 The Power Method, Convergence

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Before we make the algorithm practical, let us examine how fast the iteration converges. This requires a few definitions regarding rates of convergence.

Definition 9.3.2.1. Convergence of a sequence of scalars. Let $\alpha_0, \alpha_1, \alpha_2, \dots \in \mathbb{R}$ be an infinite sequence of scalars. Then α_k is said to converge to α if

$$\lim_{k \rightarrow \infty} |\alpha_k - \alpha| = 0.$$

Definition 9.3.2.2. Convergence of a sequence of vectors. Let $x_0, x_1, x_2, \dots \in \mathbb{C}^m$ be an infinite sequence of vectors. Then x_k is said to converge to x if for any norm $\|\cdot\|$

$$\lim_{k \rightarrow \infty} \|x_k - x\| = 0.$$

Because of the equivalence of norms, discussed in [Subsection 1.2.6](#), if a sequence of vectors converges in one norm, then it converges in all norms.

Definition 9.3.2.3. Rate of convergence. Let $\alpha_0, \alpha_1, \alpha_2, \dots \in \mathbb{R}$ be an infinite sequence of scalars that converges to α . Then

- α_k is said to converge linearly to α if for sufficiently large k

$$|\alpha_{k+1} - \alpha| \leq C|\alpha_k - \alpha|$$

for some constant $C < 1$. In other words, if

$$\lim_{k \rightarrow \infty} \frac{|\alpha_{k+1} - \alpha|}{|\alpha_k - \alpha|} = C < 1.$$

- α_k is said to converge superlinearly to α if for sufficiently large k

$$|\alpha_{k+1} - \alpha| \leq C_k|\alpha_k - \alpha|$$

with $C_k \rightarrow 0$. In other words, if

$$\lim_{k \rightarrow \infty} \frac{|\alpha_{k+1} - \alpha|}{|\alpha_k - \alpha|} = 0.$$

- α_k is said to converge quadratically to α if for sufficiently large k

$$|\alpha_{k+1} - \alpha| \leq C|\alpha_k - \alpha|^2$$

for some constant C . In other words, if

$$\lim_{k \rightarrow \infty} \frac{|\alpha_{k+1} - \alpha|}{|\alpha_k - \alpha|^2} = C.$$

- α_k is said to converge cubically to α if for large enough k

$$|\alpha_{k+1} - \alpha| \leq C|\alpha_k - \alpha|^3$$

for some constant C . In other words, if

$$\lim_{k \rightarrow \infty} \frac{|\alpha_{k+1} - \alpha|}{|\alpha_k - \alpha|^3} = C.$$

Linear convergence can be slow. Let's say that for $k \geq K$ we observe that

$$|\alpha_{k+1} - \alpha| \leq C|\alpha_k - \alpha|.$$

Then, clearly, $|\alpha_{k+n} - \alpha| \leq C^n|\alpha_k - \alpha|$. If $C = 0.99$, progress may be very, very slow. If $|\alpha_k - \alpha| = 1$, then

$$\begin{aligned}
|\alpha_{k+1} - \alpha| &\leq 0.99000 \\
|\alpha_{k+2} - \alpha| &\leq 0.98010 \\
|\alpha_{k+3} - \alpha| &\leq 0.97030 \\
|\alpha_{k+4} - \alpha| &\leq 0.96060 \\
|\alpha_{k+5} - \alpha| &\leq 0.95099 \\
|\alpha_{k+6} - \alpha| &\leq 0.94148 \\
|\alpha_{k+7} - \alpha| &\leq 0.93206 \\
|\alpha_{k+8} - \alpha| &\leq 0.92274 \\
|\alpha_{k+9} - \alpha| &\leq 0.91351
\end{aligned}$$

Quadratic convergence is fast. Now

$$\begin{aligned}
|\alpha_{k+1} - \alpha| &\leq C|\alpha_k - \alpha|^2 \\
|\alpha_{k+2} - \alpha| &\leq C|\alpha_{k+1} - \alpha|^2 \leq C(C|\alpha_k - \alpha|^2)^2 = C^3|\alpha_k - \alpha|^4 \\
|\alpha_{k+3} - \alpha| &\leq C|\alpha_{k+2} - \alpha|^2 \leq C(C^3|\alpha_k - \alpha|^4)^2 = C^7|\alpha_k - \alpha|^8 \\
&\vdots \\
|\alpha_{k+n} - \alpha| &\leq C^{2^n-1}|\alpha_k - \alpha|^{2^n}
\end{aligned}$$

Even if $C = 0.99$ and $|\alpha_k - \alpha| = 1$, then

$$\begin{aligned}
|\alpha_{k+1} - \alpha| &\leq 0.99000 \\
|\alpha_{k+2} - \alpha| &\leq 0.970299 \\
|\alpha_{k+3} - \alpha| &\leq 0.932065 \\
|\alpha_{k+4} - \alpha| &\leq 0.860058 \\
|\alpha_{k+5} - \alpha| &\leq 0.732303 \\
|\alpha_{k+6} - \alpha| &\leq 0.530905 \\
|\alpha_{k+7} - \alpha| &\leq 0.279042 \\
|\alpha_{k+8} - \alpha| &\leq 0.077085 \\
|\alpha_{k+9} - \alpha| &\leq 0.005882 \\
|\alpha_{k+10} - \alpha| &\leq 0.000034
\end{aligned}$$

If we consider α the correct result then, eventually, the number of correct digits roughly doubles in each iteration. This can be explained as follows: If $|\alpha_k - \alpha| < 1$, then the number of correct decimal digits is given by

$$-\log_{10} |\alpha_k - \alpha|.$$

Since $\log_{10}$ is a monotonically increasing function,

$$\begin{aligned}
&\log_{10} |\alpha_{k+1} - \alpha| \\
&\leq \\
&\log_{10} C|\alpha_k - \alpha|^2 \\
&= \\
&\log_{10}(C) + 2\log_{10} |\alpha_k - \alpha| \\
&\leq \\
&2\log_{10} |\alpha_k - \alpha|
\end{aligned}$$

and hence

$$\underbrace{-\log_{10} |\alpha_{k+1} - \alpha|}_{\text{number of correct digits in } \alpha_{k+1}} \geq 2 \left(\underbrace{-\log_{10} |\alpha_k - \alpha|}_{\text{number of correct digits in } \alpha_k} \right).$$

Cubic convergence is dizzyingly fast: Eventually the number of correct digits triples from one iteration to the next.

For our analysis for the convergence of the Power Method, we define a convenient norm.

Homework 9.3.2.1. Let $X \in \mathbb{C}^{m \times m}$ be nonsingular. Define $\|\cdot\|_{X^{-1}} : \mathbb{C}^m \rightarrow \mathbb{R}$ by $\|y\|_{X^{-1}} = \|X^{-1}y\|$ for some given norm $\|\cdot\| : \mathbb{C}^m \rightarrow \mathbb{R}$.

ALWAYS/SOMETIMES/NEVER: $\|\cdot\|_{X^{-1}}$ is a norm.

▼ [Solution](#)

We need to show that

- If $y \neq 0$ then $\|y\|_{X^{-1}} > 0$:

Let $y \neq 0$ and $z = X^{-1}y$. Then $z \neq 0$ since X is nonsingular. Hence

$$\|y\|_{X^{-1}} = \|X^{-1}y\| = \|z\| > 0.$$

- If $\alpha \in \mathbb{C}$ and $y \in \mathbb{C}^m$ then $\|\alpha y\|_{X^{-1}} = |\alpha| \|y\|_{X^{-1}}$:

$$\|\alpha y\|_{X^{-1}} = \|X^{-1}(\alpha y)\| = \|\alpha X^{-1}y\| = |\alpha| \|X^{-1}y\| = |\alpha| \|y\|_{X^{-1}}.$$

- If $x, y \in \mathbb{C}^m$ then $\|x + y\|_{X^{-1}} \leq \|x\|_{X^{-1}} + \|y\|_{X^{-1}}$:

$$\begin{aligned}
&\|x + y\|_{X^{-1}} \\
&= \\
&\|X^{-1}(x + y)\| \\
&= \\
&\|X^{-1}x + X^{-1}y\| \\
&\leq \\
&\|X^{-1}x\| + \|X^{-1}y\| \\
&= \\
&\|x\|_{X^{-1}} + \|y\|_{X^{-1}}.
\end{aligned}$$

What do we learn from this exercise? Recall that a vector z can alternatively be written as $X(X^{-1}z)$ so that the vector $\hat{z} = X^{-1}z$ tells you how to represent the vector z in the basis given by the columns of X . What the exercise tells us is that

if we measure a vector by applying a known norm in a new basis, then that is also a norm.

With this insight, we can perform our convergence analysis:

$$\begin{aligned} v^{(k)} - \psi_0 x_0 &= \\ A^k v^{(0)} / \lambda_0^k - \psi_0 x_0 &= \\ X \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & \left(\frac{\lambda_1}{\lambda_0}\right)^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \left(\frac{\lambda_{m-1}}{\lambda_0}\right)^k \end{pmatrix} X^{-1} v^{(0)} - \psi_0 x_0 &= \\ X \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & \left(\frac{\lambda_1}{\lambda_0}\right)^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \left(\frac{\lambda_{m-1}}{\lambda_0}\right)^k \end{pmatrix} y - \psi_0 x_0 &= \\ X \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & \left(\frac{\lambda_1}{\lambda_0}\right)^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \left(\frac{\lambda_{m-1}}{\lambda_0}\right)^k \end{pmatrix} y & \end{aligned}$$

Hence

$$X^{-1}(v^{(k)} - \psi_0 x_0) = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & \left(\frac{\lambda_1}{\lambda_0}\right)^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \left(\frac{\lambda_{m-1}}{\lambda_0}\right)^k \end{pmatrix} y$$

and

$$X^{-1}(v^{(k+1)} - \psi_0 x_0) = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & \frac{\lambda_1}{\lambda_0} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{\lambda_{m-1}}{\lambda_0} \end{pmatrix} X^{-1}(v^{(k)} - \psi_0 x_0).$$

Now, let $\|\cdot\|$ be a p-norm and $\|\cdot\|_{X^{-1}}$ as defined in [Homework 9.3.2.1](#). Then

$$\begin{aligned} \|v^{(k+1)} - \psi_0 x_0\|_{X^{-1}} &= \|X^{-1}(v^{(k+1)} - \psi_0 x_0)\| \\ &= \left\| \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & \frac{\lambda_1}{\lambda_0} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{\lambda_{m-1}}{\lambda_0} \end{pmatrix} X^{-1}(v^{(k)} - \psi_0 x_0) \right\| \\ &\leq \left| \frac{\lambda_1}{\lambda_0} \right| \|X^{-1}(v^{(k)} - \psi_0 x_0)\| = \left| \frac{\lambda_1}{\lambda_0} \right| \|v^{(k)} - \psi_0 x_0\|_{X^{-1}}. \end{aligned}$$

This shows that, in this norm, the convergence of $v^{(k)}$ to $\psi_0 x_0$ is linear: The difference between current approximation, $v^{(k)}$, and the eventual vector in the direction of the desired eigenvector, $\psi_0 x_0$, is reduced by at least a constant factor in each iteration.

Now, what if

$$|\lambda_0| = \cdots = |\lambda_{n-1}| > |\lambda_n| \geq \cdots \geq |\lambda_{m-1}|?$$

By extending the above analysis one can easily show that $v^{(k)}$ will converge to a vector in the subspace spanned by the eigenvectors associated with $\lambda_0, \dots, \lambda_{n-1}$.

An important special case is when $n = 2$: if A is real-valued then λ_0 may be complex-valued in which case its conjugate, $\bar{\lambda}_0$, is also an eigenvalue and hence has the same magnitude as λ_0 . We deduce that $v^{(k)}$ will always be in the space spanned by the eigenvectors corresponding to λ_0 and $\bar{\lambda}_0$.